Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years building **production AI workflows**, **API integrations**, customer-facing systems, and reliable multi-system automation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Structures

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Translated ambiguous stakeholder workflows into FastAPI services with explicit job states, failure isolation, and monitoring, processing 1K+ AI jobs at 3x throughput through **end-to-end technical ownership directly in production**.
- Architected multi-tenant Azure SQL and cloud-storage integrations for 500+ users, reducing dashboard latency 71% through **production debugging, indexed queries, data validation, and maintainable system boundaries across workflows**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered React and Flask workflows for uploads, model controls, history, and feedback, supporting 10K+ daily requests through **REST APIs, iterative customer requirements, and production issue resolution with instructors**.
- Engineered concurrent Python ingestion with OCR, timeouts, and multimodal model fallbacks, processing 1K+ files per batch while reducing turnaround 60% through **reproducible debugging and behavioral validation at peak load**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized distributed APIs and data integrations, improving response times 30% through object-oriented patterns, AWS Glue, S3, caching, and **root-cause analysis across cloud and cross-platform environments during production releases**.
- Automated builds and 20+ unit and integration suites across Windows, Linux, and macOS using Jenkins, PyTest, Mockito, and virtual machines, strengthening **CI/CD, regression coverage, and release reliability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, React interfaces, and real-time multi-system workflows across distributed production environments**.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while applying unit tests, Sentry, Azure Monitor, and Kibana for **instrumentation, troubleshooting, and dependable production operations across services**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into technical specifications for 12+ integrations, mapping booking, payment, coupon, and status workflows before implementation to **resolve ambiguity upfront and reduce design errors 25%**.
- Built Flask APIs for cross-platform booking and payment workflows, supporting 50K+ records with sub-200-millisecond responses while providing **technical demonstrations, integration guidance, and proactive stakeholder updates during client reviews**.

Skills

Core: Python, TypeScript, JavaScript, Go, Java, SQL, React, HTML5, CSS3, Linux
AI & Agents: LLMs, Prompt Design, Tool Calling, Orchestration, RAG, Embeddings, Evaluation, Guardrails, AI Coding Tools
Integrations: REST APIs, Webhooks, CRM, Scheduling, Event-Driven Systems, Idempotency, API Design, JSON, PostgreSQL, Redis
Reliability: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git, Tracing, Observability, Logging, Feature Flags, Replay Testing, Unit Testing, Integration Testing, Debugging, Root-Cause Analysis

Projects

CallTrace Lab | *Python, TypeScript, FastAPI, React, OpenTelemetry, PostgreSQL*

- Designed a **voice-agent replay and debugging harness** that captures spans, tool-call payloads, prompts, logs, and outcomes, then replays interactions against behavioral tests using live data before feature-flagged production rollouts.

CRM Bridge | *Go, Python, REST APIs, Webhooks, PostgreSQL, Redis, Docker*

- Built an **idempotent CRM and scheduling integration layer** with webhook ingestion, field mapping, retries, reconciliation, dead-letter handling, and reusable connectors for reliable event-driven customer workflow automation across clients.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **semantic retrieval and RAG platform** over 50K+ documents using LLaMA-2, FAISS, and Pinecone, enabling contextual question answering and summarization with sub-second retrieval through optimized embedding pipelines.